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Ultrasound guided individualized PEEP reduces postoperative atelectasis in elderly patients undergoing laparoscopic radical rectal cancer surgery

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Older patients undergoing laparoscopic radical rectal cancer surgery under general anesthesia with mechanical ventilation face an increased risk of postoperative pulmonary atelectasis due to the Trendelenburg position and pneumoperitoneum. This study aims to assess whether lung ultrasound-guided individualized positive end-expiratory pressure(PEEP) ventilation can reduce postoperative atelectasis in older patients. Forty patients aged > 65 years scheduled for elective laparoscopic radical rectal cancer surgery were randomly assigned to two groups: the ultrasound-guided group (Group P) received individualized PEEP titrated by lung ultrasound, and the control group (Group C) maintained a fixed PEEP of 5 cmH_2O. PEEP was maintained until extubation in both groups. Post-extubation, lung ultrasound assessed 12 regions in both lungs. Ultrasound-guided individualized PEEP values varied significantly between individuals [median (IQR): 11 (7-11.75)]. Compared with the PEEP with a fixed 5 cmH_2O, the incidence of postoperative pulmonary atelectasis (postoperative day 1: 0 vs. 25%; $P = 0.047$) and severity [lung ultrasound score (LUS):8.5 (6-9.75) vs. 12.5 (10-13.75); $P < 0.001$)] were lower in the patients undergoing the lung ultrasound PEEP titration strategy. Meanwhile, the intraoperative drive pressure(ΔP) (6.5 ± 2.8 vs. 10.4 ± 5.8; $P = 0.01$) and the incidence of postoperative pulmonary complications(PPCs) (5% vs. 35%; $P = 0.044$) were lower in the ultrasound-guided group, and intraoperative oxygenation index(OI) (461.5 ± 39.5 vs. 415.7 ± 69.1; $P = 0.014$) and dynamic compliance(Cdyn) (36.4 ± 8.2 vs. 25.8 ± 8.9; $P < 0.001$) were elevated. The perioperative hemodynamic characteristics were comparable between the two groups. Lung ultrasound-guided individualized PEEP decreased the incidence and severity of postoperative atelectasis in older patients undergoing laparoscopic rectal cancer surgery. This strategy improved intraoperative respiratory mechanics (Cdyn, ΔP, OI) and reduced PPCs without hemodynamic compromise.

Trial registration **This clinical trial was registered at the Chinese Clinical Trial Registry (Registration No.:ChiCTR2300078385, 07/12/2023, www.chictr.org.cn).**

Keywords Lung ultrasound, Individualized PEEP, Optimal PEEP, Pulmonary atelectasis, Laparoscopic surgery: older patients

Abbreviations

PEEP	Positive end-expiratory pressure ventilation
PPCs	Postoperative pulmonary complications
ΔP	Driving pressure
VT	Tidal volume
SpO_2	Oxygen saturation
BMI	Body mass index

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MAP	Mean arterial pressure
HR	Heart rate
ECG	Electrocardiography
$P_{ET}CO_2$	End-tidal carbon dioxide pressure
FiO_2	Oxygen concentration
PACU	Post-anesthesia care unit
COPD	Chronic obstructive pulmonary disease
P_{plat}	Platform pressure
OI	Oxygenation index
LPLV	Lung protective lung ventilation
Cdyn	Dynamic compliance
LUS	Lung ultrasound score
Cstat	static compliance
EELV	end-expiratory lung volume
EIT	Electrical impedance tomography
V_A/V_Q	Alveolar ventilation/ ventilation-perfusion
RM	Recruitment maneuvers

Patients receiving mechanical ventilation under general anesthesia frequently develop perioperative atelectasis[1]. Following anesthesia induction, diaphragmatic relaxation and cephalad displacement lead to maximally elevated pleural pressure in lung-dependent regions[2]. With 100% oxygen ventilation, 90% of anesthetized patients develop absorption atelectasis. Even at an oxygen concentration of 40%,the incidence of postoperative atelectasis reaches 57% in laparoscopic surgeries[3]. Older patients suffer from more severe postoperative atelectasis, which impairs gas exchange and oxygenation[4].Furthermore, pneumoperitoneum and Trendelenburg positioning during laparoscopic procedures increase intra-abdominal pressure and reduce functional residual capacity, exacerbating compression atelectasis and respiratory dysfunction. These factors collectively prolong hospital stays and increase pulmonary complication rates[5,6].

Protective ventilation strategies, widely used in critical care and general anesthesia, rely on PEEP as a key component to improve perioperative ventilation and oxygenation[7]. However, optimizing PEEP to prevent atelectasis while avoiding hyperinflation remains challenging. While studies suggest individualized PEEP titration using ΔP, lung compliance, or electrical impedance tomography(EIT) may be beneficial[8,9] these methods face limitations due to imprecise data or equipment complexity.

In recent years, lung ultrasound has been widely adopted for assessing PPCs and monitoring ventilation changes during mechanical ventilation under general anesthesia. Its advantages in detecting perioperative atelectasis, guiding individualized PEEP titration, and improving intraoperative pulmonary ventilation have become increasingly evident[10–12]. For diagnosing pulmonary pathologies, lung ultrasound demonstrates superior accuracy compared to chest X-rays and comparable performance to CT scans[13].

Current evidence remains inconclusive regarding the impact of lung ultrasound-guided individualized PEEP on postoperative atelectasis in older patients undergoing laparoscopic radical resection for rectal cancer. This study aims to optimize intraoperative respiratory mechanics and reduce both the incidence and severity of postoperative atelectasis through lung ultrasound-guided PEEP titration in this older patients.

Methods

This randomized controlled trial, approved by the Ethics Committee of the Affiliated Hospital of North Sichuan Medical College (Approval No. 2023ER282-1), was conducted at the Affiliated Hospital of North Sichuan Medical College. The study enrolled its first participant on July 21, 2023, and was retrospectively registered with the Chinese Clinical Trial Registry (Registration No: ChiCTR2300078385, 07/12/2023,). Written informed consent was obtained from all participants, and all study procedures were performed in strict accordance with the Consolidated Standards of Reporting Trials (CONSORT) guidelines.

Grouping and treatment

This randomized controlled single-blind trial employed computer-generated randomization (1:1 allocation ratio) to assign patients to either Ultrasound-guided individualized PEEP (Group P) or Fixed PEEP (Group C).Independent staff prepared sequentially numbered, sealed opaque envelopes containing group assignments. Upon operating room admission, patients were randomly assigned through envelope selection. PEEP titration and anesthesia management were conducted by the attending anesthesiologist, while ultrasound evaluators, follow-up physicians, and data analysts were blinded to group assignment.

Patients

A total of 40 older patients aged > 65 years with a Body mass index(BMI) between 18 and 24 kg/m^2, ASA grades II–III, and scheduled for laparoscopic radical rectal cancer surgery were included in this study. Exclusion criteria were: chest deformity, pneumothorax, massive pleural effusion, acute lung infection, chronic obstructive pulmonary disease, asthma, severe hepatic and renal dysfunction, cardiovascular and cerebral vascular diseases, anemia, and patients who underwent a second postoperative surgery.

Anesthesia

After entering the operating room, the patient's vital signs—oxygen saturation (SpO_2), mean arterial pressure (MAP), heart rate (HR), and three-lead electrocardiography (ECG)—were continuously monitored. A radial artery was cannulated under local anesthesia (2% lidocaine, 1 ml) to allow for invasive blood pressure monitoring

and blood gas analysis. Following pre-oxygenation, the patient received sequential intravenous injections of midazolam 0.02 mg/kg, sufentanil 0.3–0.5 ug/kg, etomidate 0.2–0.4 mg/kg, and cisatracurium 0.15 mg/kg for anesthesia induction. Tracheal intubation was then performed under visual laryngoscopy, and mechanical ventilation was initiated with tidal volume (VT) 6–8 mL/kg, oxygen concentration (FiO_2) 50%, respiratory rate (RR) 12 breaths/min, inspiratory-expiratory ratio (I: E) 1:2, and end-inspiratory pause 20%. Intraoperative end-tidal carbon dioxide pressure ($PETCO_2$) was maintained at 35–45 mmHg. Intraoperative anesthesia was maintained using a combination of intravenous and inhalational agents, with drug dosages adjusted based on hemodynamic changes. After pneumoperitoneum was initiated, the patient was placed in a 45° head-down-foot-up position (Trendelenburg position), which was maintained until the end of surgery. Muscle relaxants were discontinued 40 min before the end of surgery, and sevoflurane was stopped 10 min prior, with an additional 10 ug of sufentanil administered intravenously. At the conclusion of the operation, the endotracheal tube was removed once the patient showed signs of awakening, and an intravenous analgesic pump was connected. The patient was then transferred to the post-anesthesia care unit (PACU), where oxygen was administered via a face mask, and vital signs were continuously monitored.

Intervention

In group C, PEEP was set at 5 cmH_2O after intubation and applied until extubation. Patients in Group P were initiated on a PEEP of 5 cmH_2O following endotracheal intubation. Five minutes after the establishment of pneumoperitoneum, lung ultrasound scanning was performed, and PEEP was subsequently adjusted. If the LUS in any region was ≥ 2, PEEP was increased by 2 cmH_2O. Lung ultrasound examination was repeated 5 min after each PEEP adjustment. This process continued until no significant atelectasis was detected (defined as LUS < 2 in either regions), at which point the prevailing PEEP level was recorded as the individualized PEEP. If no significant lung collapse was detected upon initiation of PEEP titration, the individualized PEEP was considered to be 5 cmH_2O. The maximum permissible PEEP was 15 cmH_2O. Titration was terminated if any of the following occurred during the procedure: Hypoxemia ($SpO_2 \leq 92\%$, or an absolute intraoperative decrease in $SpO_2 > 5\%$); Hypotension (MAP decreased > 20% compared to the preoperative baseline value, persisting for > 3 min, with poor responsiveness to fluid resuscitation or vasoactive agents); Peak airway pressure exceeding 40 cmH_2O; Development of life-threatening severe ventilation or surgery-related complications (e.g., pneumothorax, severe subcutaneous emphysema).

Measurements

Lung ultrasound examinations were performed twice for each patient: 5 min before endotracheal intubation and 15 min after extubation. All ultrasound scans were conducted by the same anesthesiologist, who was blinded to group allocation and trained in adult lung ultrasonography.

The assessor divided each side of the thorax into 6 regions using 3 longitudinal (parasternal, anterior and posterior axillary) and 2 transverse (above the diaphragm, 1 cm below the nipple) lines. The 12 lung regions were scanned sequentially from right to left, head to tail, and front to back. Each region was evaluated with a two-dimensional view of the probe placed parallel to the ribs. Each region was scored between 0 and 3 based on ultrasound scoring of lung solidity and ventilation. (0) fewer than 3 isolated B-lines; (1) multiple well-defined B-lines, ≥ 3; (2)Multiple coalescent B-lines with thickened or fragmented pleural line, and small subpleural consolidations; (3) "White lung" pattern or large subpleural consolidations[14]. Atelectasis was defined as LUS ≥ 2 in any region[14,15]. The sum of the ultrasound scores from all 12 lung regions was calculated as the final LUS. A higher score indicated more severe impairment in ventilation[16] (Fig. 1).

Outcomes

The primary outcome was post extubation LUS. Secondary outcomes included respiratory mechanics parameters: peak pressure (P_{peak}), platform pressure (P_{plat}), Cdyn, and ΔP; hemodynamic parameters: HR, MAP, incidence of intraoperative hypotension, and rate of vasoactive drug use; intraoperative blood gas parameters: OI, PH, $PaCO_2$, HCO^{3-}; and 3-day postoperative PPCs[17]. The secondary outcomes were measured at the following times: T_0(at admission), T_1(5 min after intubation), T_2(30 min after pneumoperitoneum), T_3(60 min after pneumoperitoneum), T_4(at the end of pneumoperitoneum and before extubation), and T_5 (15 min after extubation). Patients were followed for 3 days postoperatively to monitor for the occurrence of PPCs. Lung ultrasound examinations were performed on the postoperative 1st and 3rd day. The diagnosis of atelectasis was based on the LUS.

Statistical analysis

Based on the pilot study results using LUS after extubation as the primary outcome (mean ± SD: 6.4 ± 2.72 for individualized PEEP vs. 8.6 ± 1.14 for fixed PEEP), we conducted a priori sample size calculation. With a power (1-β) of 0.8 to avoid type II errors and a two-sided α level of 0.05, PASS software indicated a minimum requirement of 32 patients (16 per group). Accounting for a 20% dropout rate, we planned to enroll 40 patients (20 per group).

Statistical analysis was performed using SPSS 26.0. Normally distributed measures were expressed as the mean (standard deviation), two independent samples t-test was used for comparison between groups, and repeated measures analysis of variance (ANOVA) was used for comparison of multiple time points. Non-normally distributed measures were expressed as median (M) and interquartile range (IQR), and the Mann-Whitney U rank sum test was used. Count data were expressed as frequency (%), and comparisons were made using the chi-square test. The continuous corrected chi-square test was used when the minimum expected number (T) was > 5, and the Fisher exact test was used when T < 5. For correlations between two variables, the Pearson correlation

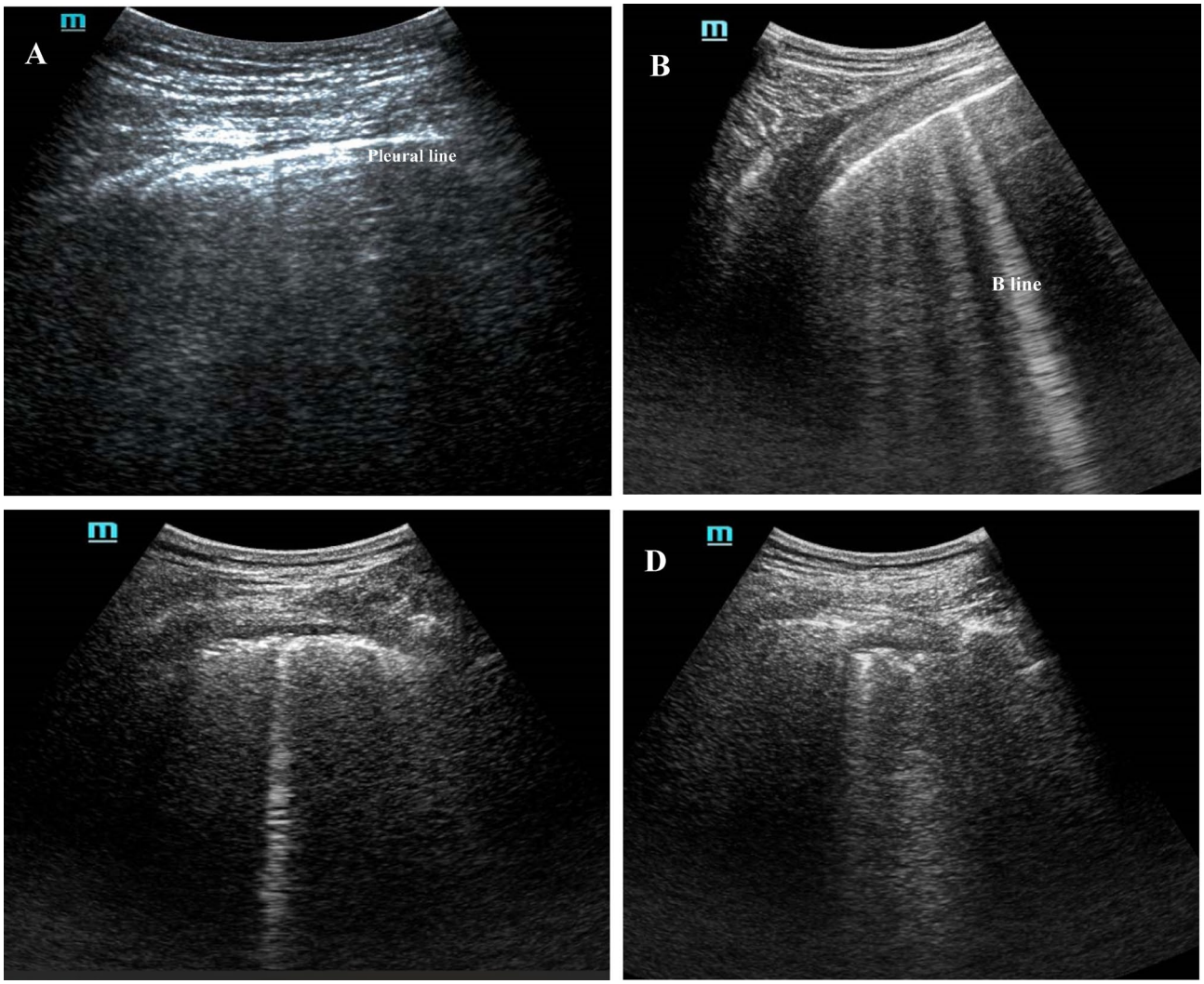


Fig. 1. LUS is assessed by subpleural solidity, B-lines, and pleural line changes. (A) LUS = 0, normal pleural lines with < 3 B-lines; (B) LUS = 1, ≥ 3 B-lines; (C) LUS = 2, presence of consolidated B-lines, thickening or breakage of pleural lines; (D) LUS = 3, subpleural solidity or pulmonary solidity.

test was used. Failure to conform to a bivariate normal distribution was tested using the Spearman test. $P < 0.05$ was considered statistically significant.

Results

Among 45 screened patients, three patient met exclusion criteria and two patients declined participation, and then 40 patients were randomly divided into two groups. All received protocol-compliant ventilation and were included in final analysis(Fig. 2).

Baseline features

A total of 40 patients were included in this study. There was no difference in gender, age, BMI, ASA classification, percentage of smoking, or prevalence of hypertension between the two groups ($P > 0.05$). The intraoperative mechanical ventilation time, pneumoperitoneum time, pneumoperitoneum pressure, Trendelenburg position time, fluid infusion, blood loss, operative time, and urine output in the two groups were comparable ($P > 0.05$), (shown in Table 1).

LUS

There was no significant difference in the LUS before intubation between group P [2 (1.25-3)] and group C: [2 (2–3)] ($P = 0.471$), (Fig. 3). Compared with Group C, Group P demonstrated significantly lower total LUS after extubation (median difference – 4; 95% CI: – 5–-2; $P < 0.001$). Significant reductions were also observed in specific regions: Left posterior upper (median difference – 1; 95% CI: – 2–0; $P = 0.003$); Right anterior lower (median difference 0; 95% CI: – 2–0; $P = 0.038$); Right lateral upper (median difference – 1;95% CI: -1–0; $P = 0.006$); Right

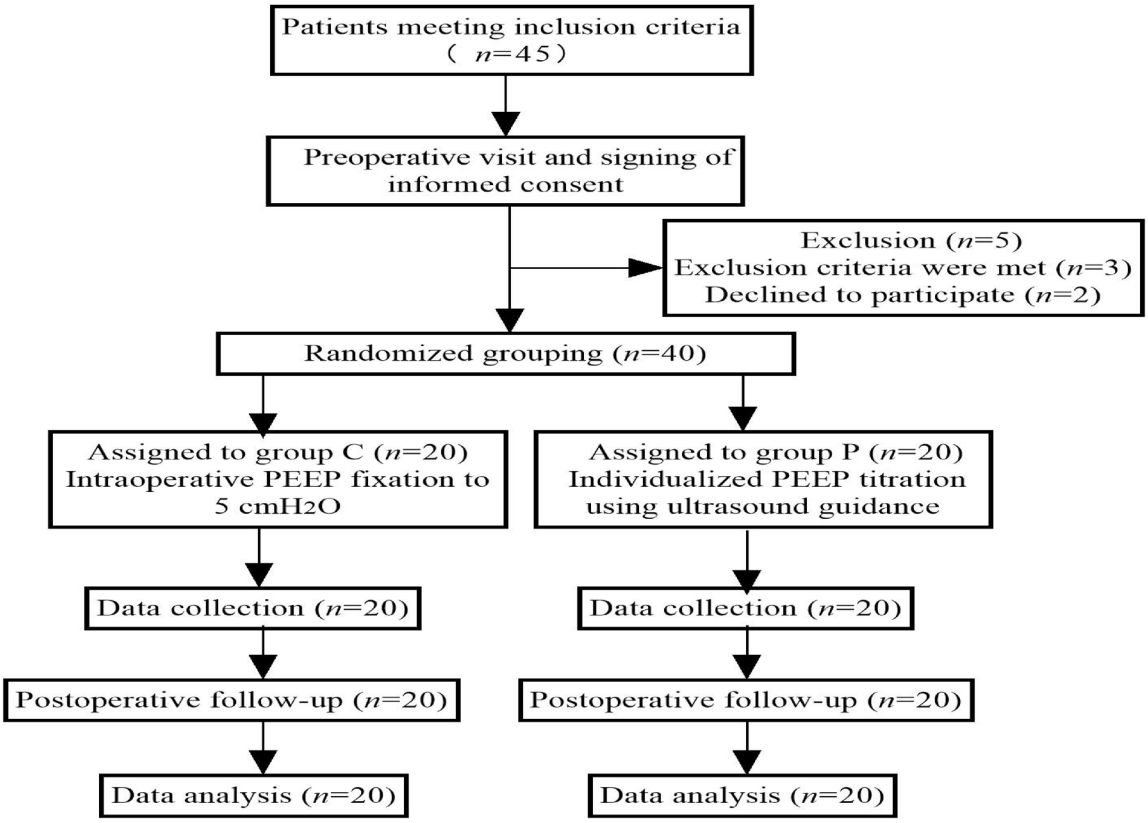


Fig. 2. Flowchart of enrolment and outcomes.

Characteristic	Group P($n=20$)	Group C($n=20$)	*P*-value
Age (yr)	71.45(4.5)	71.25(5.4)	0.899
Male [n (%)]	11(55)	9(45)	0.752
BMI(kg/m^2)	23.34(2.6)	22.75(2.9)	0.498
ASA II/III(n)	8/12	7/13	0.752
Smoking status [n (%)]	3(15)	4(20)	1.0
Hypertension[n (%)]	6(30)	9(45)	0.327
Operative time (min)	184.6(42.7)	195.2(38.9)	0.417
Mechanical ventilation time (min)	243.5(43)	254.8(42.3)	0.409
Pneumoperitoneum time (min)	158.5(41.7)	175(38.9)	0.204
Trendelenburg position time (min)	138.5(38)	159.4(38.1)	0.09
Pneumoperitoneum pressure(mmHg)	12.5(0.76)	12.7(0.92)	0.459
Blood loss(ml)	114.5(122.7)	93(108.9)	0.57
Urine output(ml)	400(234)	520(327)	0.19
fluid infusion(ml)	*	*	*
Crystalline fluids(ml)	1445(250)	1465(357.3)	0.839
Colloidal fluids(ml)	470(141.8)	475(200)	0.928

Table 1. Baseline patient characteristics. Entries are the mean (SD) or numbers (%). BMI: Body Mass Index; ASA: American Society of Anesthesiologists; Group P: the ultrasound-guided group; Group C: the control group;

lateral lower (median difference – 1; 95% CI: -1–-1; $P=0.001$).No significant intergroup differences were found in other lung regions ($P>0.05$), (shown in Table 2).

Compared with Group C, Group P exhibited significantly lower post-extubation LUS across all pulmonary regions: Anterior: median difference – 2 (95% CI: -2– 0; $P=0.005$) ;Lateral: median difference – 2 (95% CI: – 2– – 1; $P<0.001$) ;Posterior: median difference – 2 (95% CI: – 3– – 1; $P=0.001$) ;Superior: median difference – 2 (95% CI: – 3– -1; $P<0.001$) ;Inferior: median difference – 2 (95% CI: – 3– -1; $P<0.001$) ;Left: median difference – 2

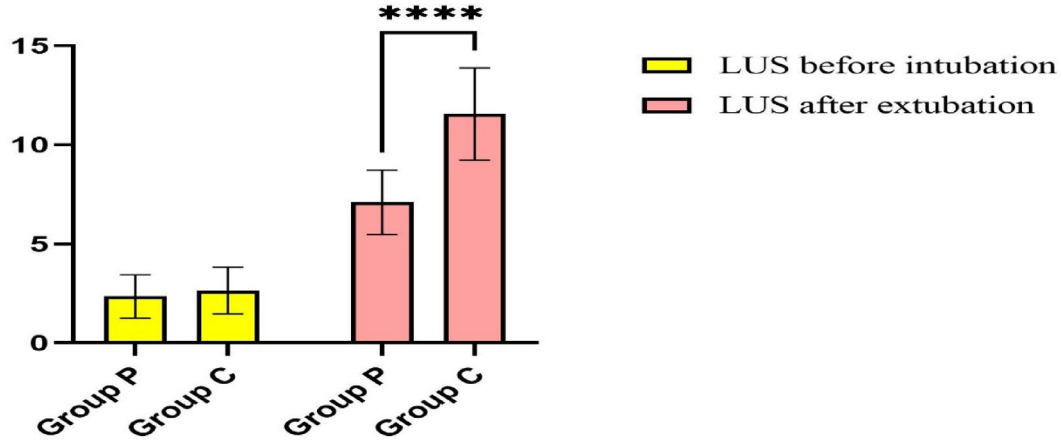


Fig. 3. Histogram comparing the LUS of the two groups [score, M (IOR)].**** represents $P < 0.0001$.

Regional LUS	Group P($n = 20$)	Group C($n = 20$)
Total	8.5(6-9.75)[a]	12.5(10-13.75)
Left front upper	0	0
Left front lower	0	0(0–1)
Left outer upper	0	0(0–1)
Left Outer lower	1(0-1.75)	1(0.25-1)
Left rear upper	0.5(0–1)[a]	2(1–2)
Left rear lower	2(2–2)	2(2–2)
Right front upper	0	0
Right front lower	0 [a]	1(0–2)
Right outer upper	0 [a]	1(0–1)
Right outer lower	0(0–1)[a]	1(1–2)
Right rear upper	1(0–2)	2(1–2)
Right rear lower	2(1–2)	2(2–2)

Table 2. Comparison of LUS after extubation. Entries are the M (IOR), LUS: lung ultrasound score; Group P: the ultrasound-guided group; Group C: the control group; [a]$P < 0.05$ vs Group C.

(95% CI: – 3– – 1; $P = 0.001$) ;Right: median difference – 3 (95% CI: -4– -2; $P < 0.001$). Group P had significantly higher LUS in some regions (lateral, posterior, superior, inferior, left, and right) compared to before intubation ($P < 0.05$), while group C had significantly higher LUS in all regions ($P < 0.05$). LUS was substantially higher in posterior lung regions in group P following extubation compared to anterior and lateral ($P < 0.001$). The inferior LUS was substantially higher than the superior LUS ($P < 0.001$), whereas there was no significant difference between the left and right lung LUS ($P = 0.769$). In group C, there was no significant difference between the left and right LUS ($P = 0.124$). However, posterior LUS was significantly higher than anterior and lateral LUS ($P < 0.001$), and inferior LUS was significantly higher than superior LUS ($P < 0.001$) (shown in Table 3). For patients over the age of 65, there was no age-related increase in LUS after extubation ($R^2 = 0.04$, $P = 0.922$).

P_{peak}, P_{plat}, cdyn, and ΔP

Comparative analysis between Group P and Group C revealed significant differences in respiratory mechanics at various time points. Group P demonstrated significantly higher P_{peak} values at T_2 ($P = 0.038$)and T_4 ($P < 0.001$). Group P showed markedly improved Cdyn during T_2–T_4, T_2: mean difference 7.54 (95% CI: 2.5–12.57; $P = 0.004$); T_3: mean difference 10.57 (95% CI: 5.1–16.04; $P < 0.001$); T_4: mean difference 6.8 (95% CI: 0.27–13.32; $P = 0.042$).Significant reductions in ΔP were observed in Group P at T_2: mean difference – 3.6 (95% CI: – 6.29– -0.9; $P = 0.01$), T_3: mean difference – 3.95 (95% CI: – 6.9– – 1.01; $P = 0.01$). There was no significant difference in P_{plat} between group P and group C at T_1–T_5 ($P > 0.05$). In comparison to T_1, Cdyn declined at T_2–T_3 and ΔP increased at T_2–T_3 in group P, while Cdyn decreased at T_2–T_4 and ΔP increased at T_2–T_4 in group C ($P < 0.05$). (shown in Table 4).

OI, PH, $PaCO_2$ and HCO^3

Compared with group C, the OI was significantly higher in group P at T_3–T_5 ($P = 0.014$, $P = 0.001$, $P = 0.001$, respectively), and PH, $PaCO_2$, and HCO^{3-} were not significantly different between the two groups at T_0–T_5 ($P > 0.05$), (shown in Table 5).

Perioperative hemodynamics

There was no significant difference in MAP and HR at T_0–T_5 between group P and group C (Figs. 4 and 5), nor was there a significant difference in the use of intraoperative vasoactive drugs or the occurrence of hypotension between the two groups ($P > 0.05$).

Regional LUS	Group	Pre-intubation	Post-extubation
Anterior	Group P	0	0(0-0.75)[a]
	Group C	0	2(0–2)[b]
Lateral	Group P	1(0–2)	2(1–2)[ab]
	Group C	1(1–2)	3(2.25-4)[b]
Posterior	Group P	1(1–2)	5(4–6)[ab]
	Group C	1(0–2)	7(6–8)[b]
Superior	Group P	0(0–1)	2(1–3)[ab]
	Group C	0	4.5(3–5)[b]
Inferior	Group P	2(1–3)	5.5(4–6)[ab]
	Group C	2(1.25-3)	7(7–8)[b]
Left	Group P	1(1–2)	3(3–4)[ab]
	Group C	1(0–2)	6(4–7)[b]
Right	Group P	1(0.25-2)	3.5(2–5)[ab]
	Group C	2(1–2)	6(5.25–7.75)[b]

Table 3. Comparison of LUS before intubation and after extubation. Entries are the M (IOR). LUS: lung ultrasound score; Group P: the ultrasound-guided group; Group C: the control group; [a]$P < 0.05$ vs. Group C, [b]$P < 0.05$ vs. pre-intubation.

Parameter	Group	T_1	T_2	T_3	T_4
$P_{peak}(cmH_2O)$	Group P	14.5(1.4)	25.5(2.2)[a]	25.1(2.5)	21.6(2.3)[a]
	Group C	15.4(1.5)	23.4(3.6)	24.1(3.6)	17.2(2.1)
$P_{plat}(cmH_2O)$	Group P	8.8(2.1)	17.1(2.3)	17.2(2.1)	11.7(2.3)
	Group C	9.9(2.3)	14.5(5.7)	15.1(5.8)	10.9(3.0)
$Cdyn(ml/cmH_2O)$	Group P	46.7(11.1)	33.9(5.8)[ab]	36.4(8.2)[ab]	45.0(10.1)[a]
	Group C	44.7(13.2)	26.4(9.5)[b]	25.8(8.9)[b]	38.3(10.3)[b]
$\Delta P(cmH_2O)$	Group P	4.3(1.9)	6.4(2.6)[ab]	6.5(2.8)[ab]	4.6(2.0)
	Group C	5.0(2.4)	10(5.3)[b]	10.4(5.8)[b]	6.1(2.9)[b]

Table 4. Comparison of ventilation indices at different time points. Entries are the mean (SD). P_{peak}: peak pressure; P_{plat}: platform pressure; Cdyn: intraoperative dynamic compliance; ΔP: driving pressure; Group P: the ultrasound-guided group; Group C: the control group; [a]$P < 0.05$ vs. Group C, [b]$P < 0.05$ vs. T_1.

Indexes	Group	T_0	T_1	T_2	T_3	T_4	T_5
OI	Group P	313.4(29.3)	417.3(51)	457.2(45.2)	453.3(48.1)[a]	437.5(63.5)[a]	365.7(40.8)[a]
	Group C	324(33.1)	403.1(77.1)	438.9(50)	417.2(70.4)	401.6(58.1)	335.2(39)
PH	Group P	7.37(0.03)	7.34(0.05)	7.27(0.06)	7.27(0.06)	7.31(0.05)	7.33(0.04)
	Group C	7.39(0.03)	7.36(0.04)	7.3(0.04)	7.29(0.04)	7.31(0.05)	7.33(0.05)
$PaCO_2$	Group P	38.2(3.0)	37.2(4.1)	46.1(6.3)	47.8(6.0)	42.8(6.4)	42.4(4.5)
	Group C	38.4(2.0)	36.5(4.3)	44.8(6.1)	45.8(6.4)	42.0(6.3)	42.1(4.5)
HCO_{3-}	Group P	22.6(2.2)	21.8(1.9)	21.2(2.0)	21.1(1.2)	21.7(2.4)	22(2.0)
	Group C	23.5(1.6)	22.3(1.8)	21.8(1.9)	21.2(1.5)	20.8(1.9)	22.5(2.5)

Table 5. Comparison of blood gas indices at different time points. Entries are the mean (SD). OI: perioperative oxygenation index; Group P: the ultrasound-guided group; Group C: the control group; [a]$P < 0.05$ vs. Group C.

PPCs and lung atelectasis

The incidence of PPCs on the third day after surgery was significantly lower in group P than in group C (5%; 95% CI: 0.001–0.249 vs. 35%; 95%CL: 0.154–0.592; $P = 0.044$). Only one patient suffered from hypoxemia in Group P. In group C, five patients were found to have pulmonary atelectasis (detected by lung ultrasound), and the other two cases of PPCs exhibited hypoxemia. The symptoms of hypoxia were all improved after oxygenation. On the first day after surgery, the incidence of pulmonary atelectasis was less in group P than in group C(0; 95% CI: 0–0.168 vs. 25%; 95%CL: 0.087–0.49; $P = 0.047$). On the third day after surgery, there was no significant difference in the incidence of pulmonary atelectasis in group P compared with group C ($P = 0.487$), (shown in

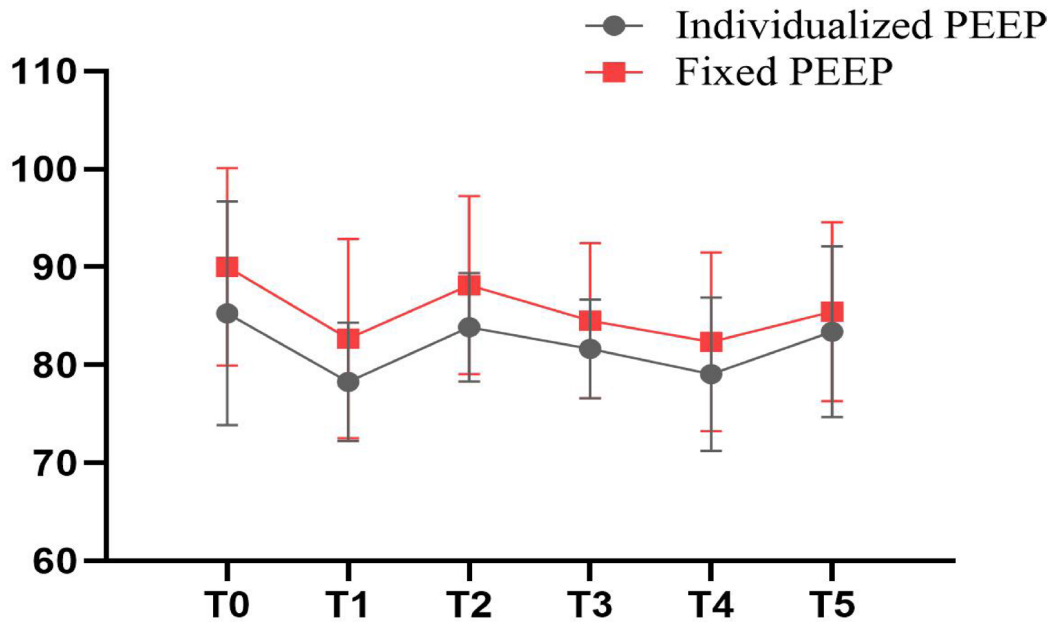


Fig. 4. Line diagram of MAP. Points represent means and whiskers 95% confidence intervals from the data at that given time point.

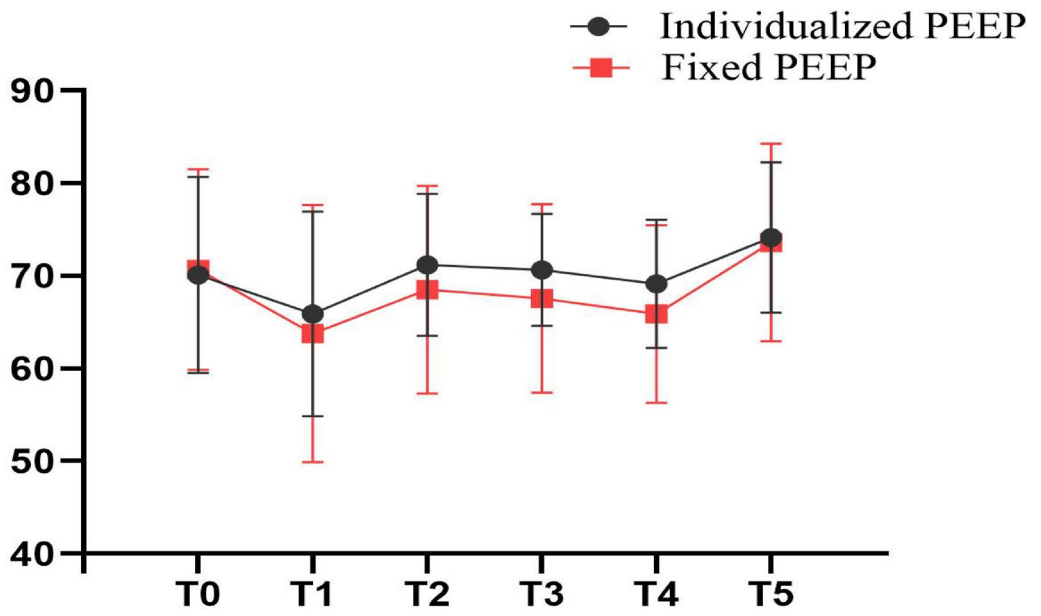


Fig. 5. Line diagram of HR. Points represent means and whiskers 95% confidence intervals from the data at that given time point.

Table 6). The optimal PEEP value in the individualized PEEP group was 11 (7–11.75) cmH_2O. For patients over 65 years old, increasing age did not correlate with an increasing PEEP value ($P = 0.666$).

Discussion

This study demonstrates that lung ultrasound-guided individualized PEEP improves postoperative atelectasis, enhances perioperative Cdyn and OI, reduces intraoperative ΔP and the incidence of PPCs, while maintaining hemodynamic stability.

Most surgical patients receiving mechanical ventilation develop postoperative atelectasis, which reduces lung compliance, impairs oxygenation, and increases pulmonary vascular resistance[2,18]. Individualized PEEP has been proposed for mechanically ventilated patients under general anesthesia[19]. The quantitative relationship between LUS and atelectasis volume measured by computed tomography demonstrated high sensitivity (87.7%),

Parameter	Time	Group *P*	Group C	*P*-value
PPCs[n (%)]	3d	1(5%)[a]	7(35%)	0.044
pulmonary atelectasis [n (%)]	1d 3d	0 [a] 0	5(25%) 2(10%)	0.047 0.487

Table 6. PPCs and postoperative pulmonary atelectasis. PPCs: postoperative pulmonary complications; Group C: the control group; [a]$P < 0.05$ vs. Group C.

specificity (92.1%), and accuracy (90.8%) in diagnosing postoperative atelectasis in anesthetized adults[20].In this study, the optimal PEEP varied substantially among individuals (7–14 cmH_2O) and showed no correlation with age. Notably, laparoscopic surgery in the Trendelenburg position required higher individualized PEEP levels 11 (7–11.75) cmH_2O compared to other surgical positions or open surgery, as this approach further exacerbates ventilation loss due to cephalad diaphragmatic shift[21,22].

This study demonstrated that ultrasound-guided individualized PEEP effectively reduced regional LUS in patients undergoing laparoscopic radical resection for rectal cancer, indicated its ability to recruit collapsed alveoli independently of body position and lung compliance. Heejoon Jeong et al.[23] reported that under pneumoperitoneum in the Trendelenburg position, pulmonary ventilation was predominantly distributed in the basal and posterior lung regions. The combined effects of pneumoperitoneum and positioning shifted ventilation toward non-dependent lung areas, resulting in increased poorly ventilated zones in gravity-dependent regions. Our data showed a significant increase in LUS in the posterior and basal lung fields after intubation in both groups. This effect was gradually reversed with increasing PEEP in the individualized group, but in the fixed PEEP group, this effect persisted until spontaneous respiration resumed. On the first day after surgery, the incidence of atelectasis was significantly lower in the individualized PEEP group. However, this difference between groups diminished over time. Mohamed Elshazly et al.[24] found no significant difference in atelectasis incidence at 24 h postoperatively between ultrasound-guided PEEP and fixed PEEP groups, which may be due to the limited sensitivity of chest radiography used for atelectasis detection in patients with hypoxemia. Existing evidence suggested that the benefits of individualized PEEP on end-expiratory lung volume (EELV) and alveolar recruitment disappear within 2–6 h after extubation, while post-extubation atelectasis may persist for up to 24 hours[25].

In this study, while the individualized PEEP group demonstrated higher P_{peak} , it exhibited significantly lower ΔP, which generally remained below the 12.5 H_2O threshold, consequently reducing the risk of PPCs[26]. A large-scale meta-analysis confirmed that PPCs were associated with high PEEP levels that induce ΔP elevation[27].Hyun-Kyu Yoo et al.[28] optimized PEEP settings through Cdyn titration for laparoscopic surgery in Trendelenburg position. Although higher PEEP levels [14 (12–18) cmH_2O] improved postoperative atelectasis, they failed to reduce PPCs incidence, attributable to the elevated intraoperative ΔP (median: 15; IQR: 13.0–17.5) observed in their individualized PEEP group. Our findings demonstrated that ultrasound-guided individualized PEEP effectively reduces the intraoperative ΔP and incidence of PPCs, and improving Cdyn. These results were consistent with those of studies utilizing EIT or ΔP-based PEEP titration strategies[29,30].

Under general anesthesia, the dispersion of ventilation/perfusion (V_A/V_Q) ratios and perfusion in low V_A/V_Q regions increases with age[31]. Concurrently, atelectasis induces intrapulmonary shunting[32]leading to impaired oxygenation. In present study, ultrasound-guided individualized PEEP improved alveolar recruitment across all lung regions, thereby optimizing VA/VQ matching and reducing right-to-left shunting. This explained the consistently higher OI observed in the individualized PEEP group from pneumoperitoneum initiation through extubation–findings that parallel results documented in obese patients[24].Although elevated PEEP levels may adversely affect hemodynamics, our individualized PEEP group exhibited no significant hemodynamic fluctuations. Previous studies employing decremental PEEP titration protocols had demonstrated that high PEEP levels combined with recruitment maneuvers (RM) invariably impact hemodynamics[33,34]. In contrast, our incremental PEEP titration approach mitigated both hemodynamic fluctuations and pulmonary overdistension. Furthermore, during laparoscopic procedures, the combined effects of Trendelenburg positioning and pneumoperitoneum increase MAP by 35%, with MAP returning to baseline upon surgical completion[35].

Strengths and limitations

This study represents the first application of lung ultrasound for PEEP optimization in older patients undergoing laparoscopic surgery. Its demonstrated efficacy, practicality, and safety suggest potential as a standardized approach for individualized PEEP titration. Our protocol included postoperative lung ultrasound assessments at 1-day and 3-day intervals, enabling evaluation of whether the beneficial effects on atelectasis persist beyond the anesthesia recovery period. However, several limitations warrant consideration: First, as a single-center randomized controlled trial with a limited sample size, the study may have been underpowered to detect significant between-group differences in 3-day atelectasis incidence. Second, post-extubation lung ultrasound scans could potentially be confounded by uncontrolled recruitment maneuvers (e.g., sighing or coughing) or patient sleep state, possibly obscuring true LUS differences. Third, the absence of intraoperative LUS measurements precludes demonstration of intraoperative alveolar recruitment effects, though studies by Gouri Mini et al. and Qi Xu et al. confirm that individualized PEEP reduces both intraoperative and post-extubation LUS[30,36]. Fourth, PEEP titration was performed in the supine position post-pneumoperitoneum without subsequent positional readjustment, potentially attenuating the full therapeutic effect during pneumoperitoneum. Fifth, while individualized PEEP improved LUS in posterior and inferior lung regions, it failed to reduce LUS in posteroinferior areas—likely due to the omission of RM. Contemporary evidence confirms that individualized

PEEP alone cannot adequately recruit dorsal alveoli near the diaphragm without RM[37,38]suggesting combined ultrasound-guided PEEP with RMs as a critical research direction. Sixth, although registered retrospectively on ClinicalTrials.gov, the study protocol received ethics approval prior to first patient enrollment and was strictly adhered to without post hoc modifications.

Conclusion

In older patients undergoing laparoscopic rectal cancer surgery, optimal PEEP under general anesthesia showed wide interindividual variability and no correlation with age in patients > 65 years. Ultrasound-guided individualized PEEP improved oxygenation, lung compliance, and driving pressure while decreasing the incidence of PPCs. Additionally, it decreased both the incidence and severity of postoperative atelectasis without significant hemodynamic effects.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available but are available from the corresponding author upon reasonable request.

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Author contributions

F, Xuechao Li, Z, Fangjun Wang . Responsible for and completes study design. F . Responsible for Patient recruitment.Z . Wei Li : Data collection.Wei Li, Jixiang Wan . Data analysis.Xuechao Li, Jixiang Wan . Supervision. Jixiang Wan, Fangjun Wang . Validation.F . Completion of the original draft.Fangjun Wang . final Verification and Revision.All authors reviewed the manuscript.

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Declarations

Ethics approval and consent to participate

The study was carried out as a randomized controlled trial at the Affiliated Hospital of North Sichuan Medical College after receiving approval from the Hospital Ethics Committee (Ethics No. 2023ER282-1).

Competing interests

The authors declare no competing interests.

Additional information

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